

all the millions of items simultaneously, thanks to superposition. It can look at the entire database of products at once, instantly evaluating which ones match your search, without having to go step by step.

This ability to evaluate multiple possibilities at the same time makes quantum computers vastly more efficient for certain tasks, especially when there are lots of potential outcomes, like searching through massive databases or solving complex problems.

Entanglement

Another important feature of quantum computing is entanglement. Quantum entanglement occurs when two qubits become linked together in such a way that the state of one qubit directly affects the state of the other, even when they are far apart. In other words, once two qubits are entangled, measuring the state of one qubit (whether it's 0 or 1) immediately reveals the state of the other qubit, regardless of the distance between them. This phenomenon is what makes quantum computers so powerful, as it allows them to perform complex calculations much faster than traditional computers.

In classical computing, each operation is processed step by step, one at a time. However, with quantum entanglement, quantum computers can process linked qubits simultaneously. This means that instead of handling one calculation at a time, quantum computers can solve multiple calculations at once, greatly speeding up the process, especially for problems that involve large amounts of data or need complex processing. For example, imagine two satellites orbiting the Earth — one above India and the other above Brazil. If these satellites were using quantum entanglement for communication, any change or measurement of data from one satellite could instantly affect the data received by the other, even though they are

thousands of kilometres apart. Unlike traditional communication, which relies on signals travelling through space at the speed of light, quantum entanglement works differently. In entanglement, the data from one satellite is linked to the other in such a way that when you measure one, you instantly know the state of the other, no matter the distance between them. This connection doesn't require signals to physically travel through space, making it much harder for anyone to intercept or interfere with the communication. The result is not only faster communication but also an ultra-secure method for transferring data. This unique feature of quantum communication allows it to solve certain problems much more efficiently and securely than traditional methods.

Quantum entanglement could play a critical role in the development of ultra-secure communications. Due to the unique behaviour of entangled qubits, any attempt to intercept information being transmitted between them would be immediately noticed. This makes quantum communication nearly impossible to hack. If an intruder tries to interfere with the data, it will disturb the quantum state, alerting the parties involved. This level of security is especially valuable for sensitive communications, such as those in government, banking, or personal exchanges.

Quantum entanglement also has the potential to significantly improve cloud computing. In traditional cloud systems, servers store and process data independently. With quantum entanglement, servers could work together in real-time, sharing information and solving complex problems more efficiently. This could significantly speed up data processing for large-scale tasks, like simulations or analysing massive datasets.

Quantum entanglement has transformative potential across various industries. In cybersecurity, it can provide instant detection of any attempts

to intercept or tamper with data, making hacking nearly impossible and enhancing digital transaction security. In healthcare, quantum computers could process vast amounts of biological data simultaneously, accelerating drug discovery and helping find cures for diseases like cancer. In business, quantum entanglement could optimise supply chains by synchronising data across different parts of a company in real-time, improving decision-making and reducing costs.

With these incredible properties—superposition and entanglement—quantum computers are pushing the boundaries of what's possible in cryptography, artificial intelligence (AI), and scientific research. Quantum computing could break current encryption methods or help create stronger, more secure systems. In AI, quantum computers could enable faster, smarter algorithms, improving areas like recommendation systems, self-driving cars, or medical diagnosis.

Recognising the importance of quantum computing, the Indian government has launched the National Quantum Mission (NQM) with a planned investment of more than Rs 60 billion from 2023-24 to 2030-31.

How did it all start?

The journey of quantum computing began over a century ago, when scientists started exploring the strange behaviour of particles like electrons and photons. This study, called quantum mechanics, laid the foundation for the development of quantum computing. Key figures such as Albert Einstein, Niels Bohr, and Erwin Schrödinger helped us understand how particles can behave in ways that don't fit with our everyday experience of physics. For example, Schrödinger created a famous thought experiment, known as 'Schrödinger's cat', to explain the idea of superposition—the concept that a quantum particle can exist in multiple states at once.